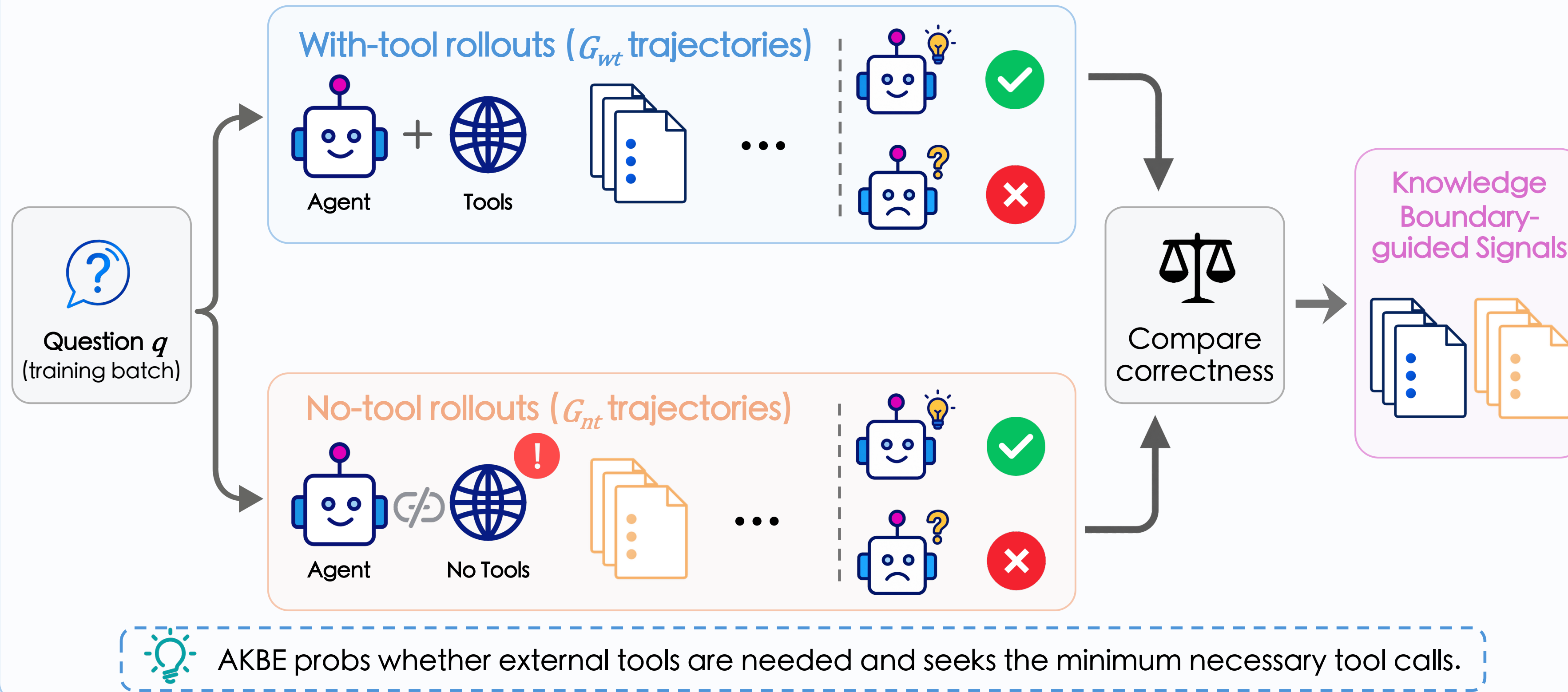
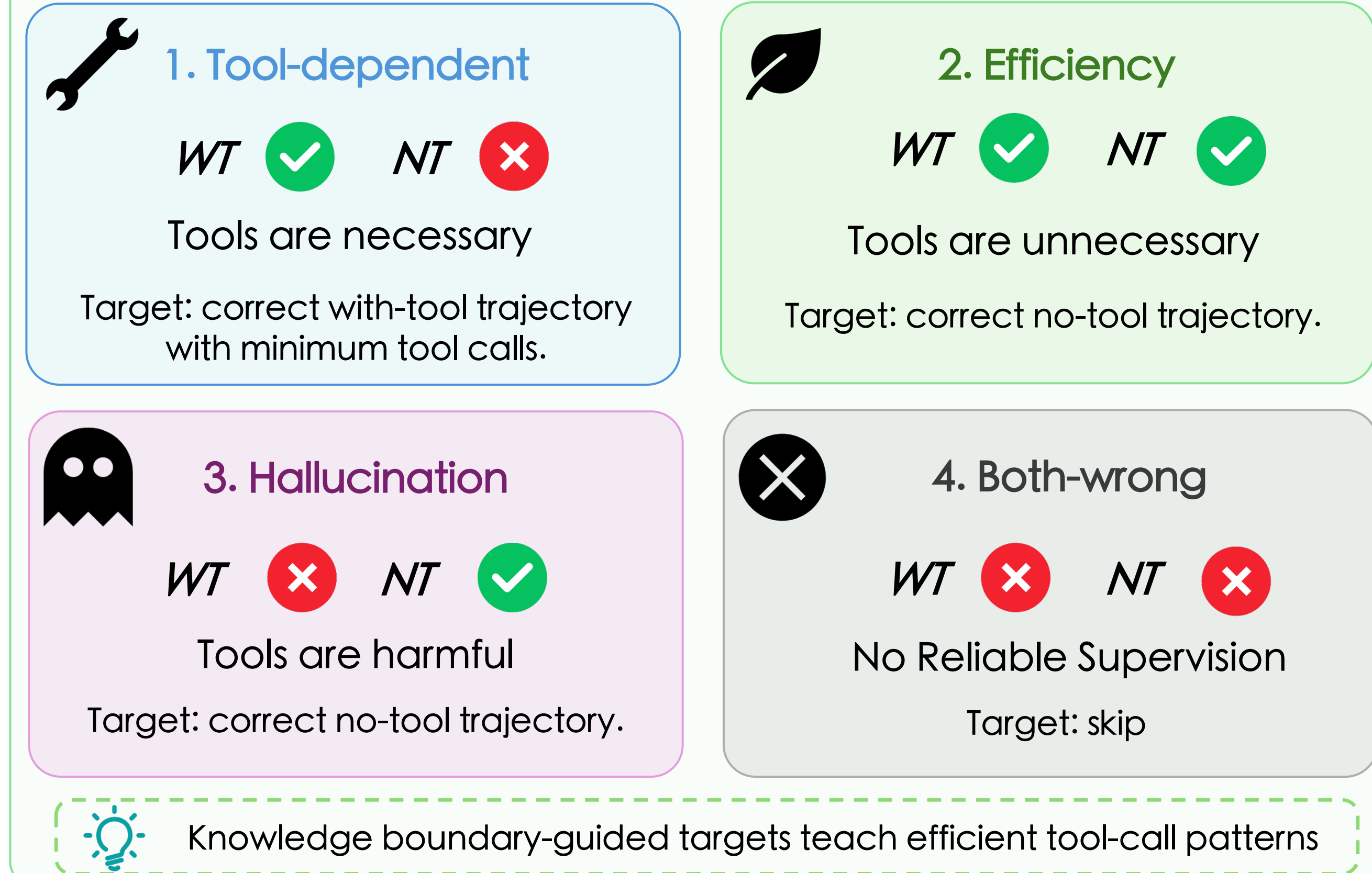


(a) Dual-Path Knowledge Boundary Probing



(b) Knowledge Boundary-guided Signal Construction



(c) Joint On-Policy Training Objective

